

The Superclouds of the Local Milky Way

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Background

Superclouds, defined as large neutral atomic hydrogen clouds with typical masses between 10^6 and $10^7 M_\odot$ and lengths on the order of kiloparsecs, were first studied by Elmegreen & Elmegreen (1983) in external galaxies. Further work was done by Elmegreen & Elmegreen (1987), who identified superclouds in the inner Milky Way. Up until now, no superclouds have been identified in the immediate solar neighborhood, raising the question if their absence is physical or the result of observational limitations in the past.

Data and Methods

We use the posterior mean of the 3D Cartesian grid dust map interpolated by Edenhofer et al. (2024) (hereafter E24), extending out to 1.25 kpc from the Sun. The map has a resolution of 2^3 pc^3 , with the data given as unitless extinction values. To convert the extinction data to hydrogen volume densities, we apply the conversion formula from O'Neill et al. (2024). For the segmentation of the dust map, the HOP structure finder adapted by Colman et al. (2024) is used.

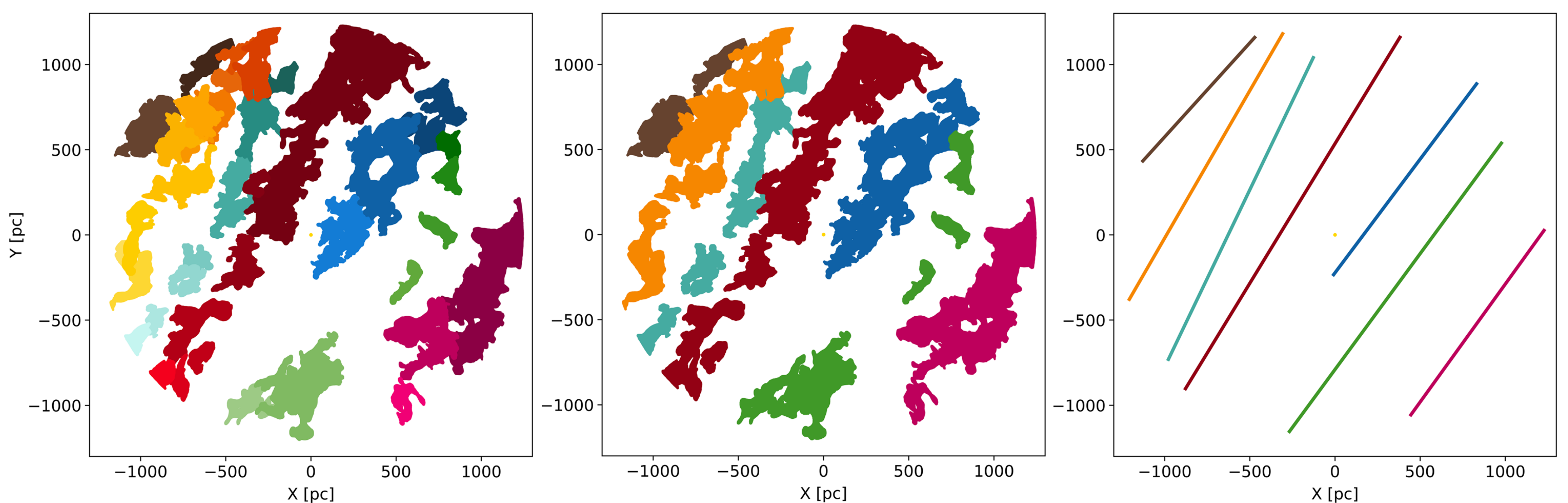


Fig. 1.: Face-on view of the 40 identified HOP clouds (left), the seven connected superclouds (middle), and the corresponding supercloud orientations (right). The superclouds show an average pitch angle of $\sim 30^\circ$, unlikely to be associated with the $\sim 10^\circ$ Local Arm. Each cloud and supercloud is assigned a distinct color, and the length of the line corresponds to the calculated length of the cloud. The Sun is represented by the yellow dot in the center, and the Galactic Center is located to the right.

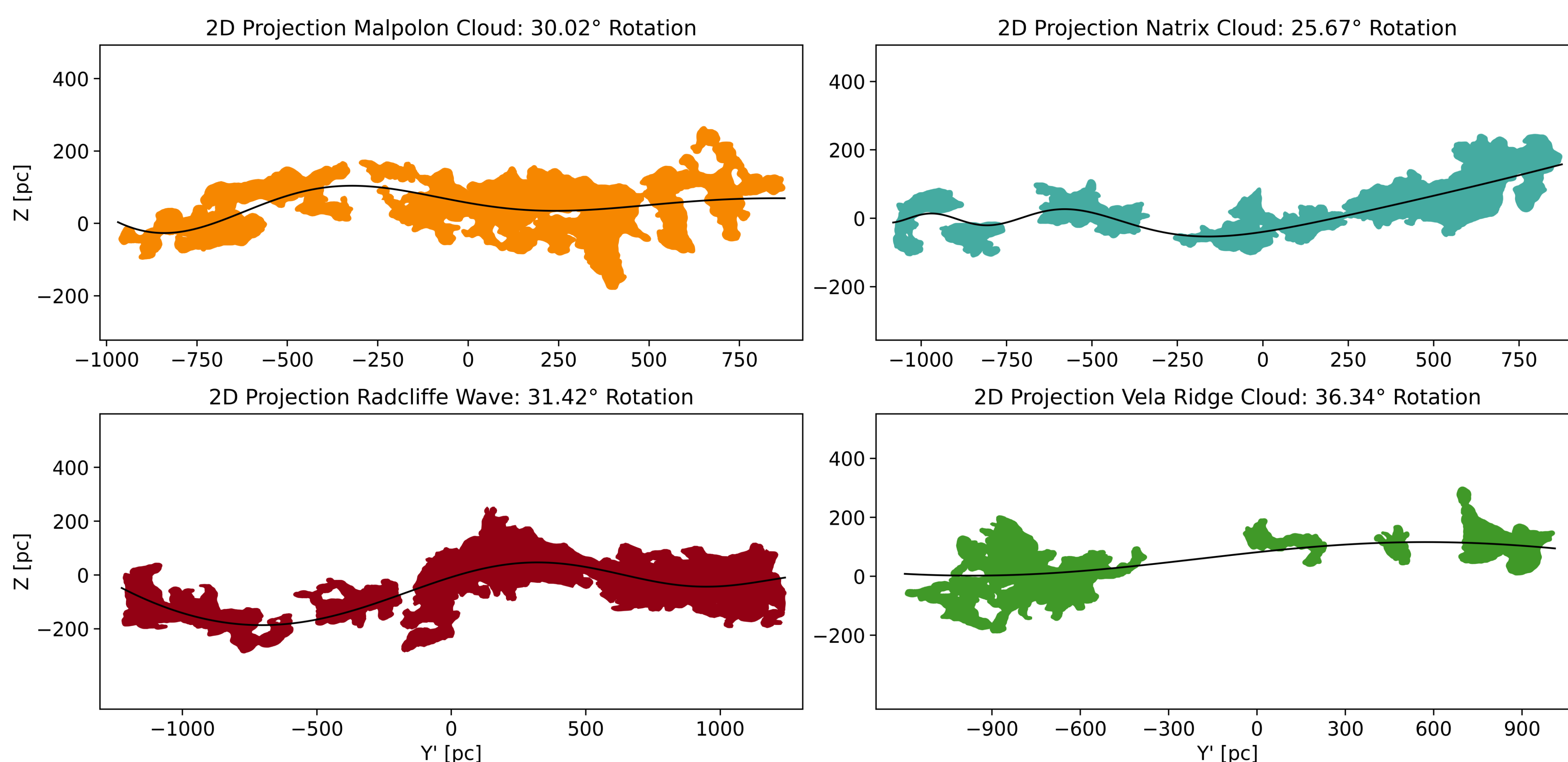


Fig. 2.: 2D projection for four out of the seven superclouds in the Z/Y' plane with a sinusoidal fit. For the fit, the clouds are rotated anticlockwise by their average face-on angle in the X/Y plane. The Malpolon cloud, Natrix cloud, and Radcliffe Wave are fitted with a damped sinusoidal, the Vela Ridge cloud is fitted with a simple sinus function.

Conclusion

We identify a distribution of seven quasi-parallel gas lanes in the E24 dust map. These structures extend over several kiloparsec, contain substantial masses on the order of $10^6 M_\odot$, and exhibit pitch angles of approximately 30° . Given their size, mass, and likely composition (predominantly atomic hydrogen), we refer to them as superclouds, following the terminology of Elmegreen & Elmegreen (1983). The discovery of a coherent network of superclouds within the local kiloparsec marks a step forward in our understanding of the Milky Way's ISM. As we refine our three-dimensional view of the nearby Galaxy, we are beginning to define a solid benchmark for connecting the complex structure of the Milky Way with the broader context of galaxy evolution and star formation in the universe.



The preprint is now available on the arXiv!

References

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